NetDoctor

NETWORK DIAGNOSTIC SYSTEM — PHASE 3 COMPLETION REPORT

Phase Completed: Phase 3 (Structured Diagnosis & Troubleshooting)	**Status:** 100% Implemented & Verified
Test Suites: 60/60 Tests Passed across 5 test suites	**Coverage:** 13 Diagnostic Steps, 11 Roadmaps, 12 Root Causes

1. Phase 3 Architecture & Implementation Summary

Phase 3 establishes the deterministic diagnostic knowledge and command interpretation engine for NetDoctor. It delivers a comprehensive Curated Networking Knowledge Base across all OSI layers, complete OS command sets for Windows, macOS, and Linux, deterministic regex output parsers, beginner-friendly plain-language translations, and comprehensive root-cause mappings.

2. Curated Diagnostic Procedures Library (13 Steps)

Step ID	Layer	Windows / macOS / Linux Command	Key Metric / Target
`step_network_adapter`	Physical/Link	`netsh interface show / networksetup / ip link`	Interface link & admin state
`step_ip_dhcp`	IP / DHCP	`ipconfig /all / ifconfig / ip addr show`	IPv4, Subnet, Gateway, APIPA
`step_gateway_ping`	Local Network	`ping 192.168.1.1 (4 packets)`	Router packet loss & latency
`step_dns_lookup`	DNS	`nslookup google.com / dig`	A/AAAA address translation
`step_internet_ping`	WAN / Internet	`ping 8.8.8.8 / 1.1.1.1`	Direct public IP routing
`step_wifi_signal`	Wireless Link	`netsh wlan show interfaces / nmcli`	SSID, Signal %, Band, Radio
`step_ping_latency_jitter`	Transport	`ping -n 10 8.8.8.8 (10 packets)`	Latency jitter & bufferbloat
`step_trace_route`	Routing Path	`tracert 8.8.8.8 / traceroute -n`	Hop 1 gateway vs ISP drops
`step_dns_flush`	DNS Cache	`ipconfig /flushdns / dscacheutil`	Clear corrupted resolver cache
`step_ip_release_renew`	DHCP Protocol	`ipconfig /release && renew`	Obtain fresh DHCP IP lease
`step_http_reachability`	Application	`curl -I https://www.google.com`	HTTP 200 OK / port response
`step_lan_peer_ping`	Local LAN	`arp -a / ip neigh show`	ARP cache & subnet discovery
`step_adapter_restart`	Interface Reset	`netsh interface set ... disable/enable`	Reinitialize network driver

3. Multi-OS Command Output Regex Interpreters

- **IP & DHCP Parsing:** Regex extractors capture IPv4 address, Subnet Mask, Default Gateway, DHCP server, DNS servers, and immediately flag APIPA addresses (169.254.x.x) and Media Disconnected states.
- **Ping & Latency Parsing:** Parses Windows, macOS, and Linux ping statistics (Packets sent/received/lost, loss %, min/max/avg RTT, TTL). Classifies Request timed out, Destination host unreachable, General failure, and evaluates latency thresholds.

- **DNS Parsing:** Analyzes nslookup, dig, and host outputs for server IP, resolved A records, and captures error signatures (DNS request timed out, NXDOMAIN, SERVFAIL).
- **Trace, Wi-Fi & HTTP Parsing:** Detects Hop 1 router timeouts, Wi-Fi signal degradation (<45%), SSID parameters, and HTTP response codes (200 OK vs 500/502/Connection Refused).

4. Curated Root Causes, Solutions & Prevention (12 Mappings)

Root Cause ID	Identified Root Cause	Verification Step
`rc_no_physical_link`	Network adapter disabled, airplane mode, or cable disconnected	`step_network_adapter`
`rc_apipa_dhcp_failure`	DHCP server lease failure resulting in self-assigned 169.254.x.x	`step_ip_dhcp`
`rc_gateway_unreachable`	Local default gateway / router unreachable or frozen	`step_gateway_ping`
`rc_dns_server_failure`	Configured DNS servers unreachable or failing name translation	`step_dns_lookup`
`rc_dns_cache_corruption`	Corrupted or stale local DNS resolver cache entries	`step_dns_lookup`
`rc_isp_wan_outage`	Local gateway reachable, but external ISP WAN link is down	`step_internet_ping`
`rc_wifi_weak_signal`	Weak Wi-Fi signal (<45%) or 2.4GHz radio interference	`step_wifi_signal`
`rc_bufferbloat_congestion`	Local bandwidth saturation and bufferbloat queuing delay	`step_ping_latency_jitter`
`rc_packet_loss_hop`	Packet drops occurring on intermediate ISP transit hop	`step_trace_route`
`rc_target_server_outage`	Target web server down or refusing connections (Internet OK)	`step_http_reachability`
`rc_lan_subnet_mismatch`	Subnet mismatch or firewall blocking local peer discovery	`step_lan_peer_ping`
`rc_adapter_driver_glitch`	Operating system network driver stack state inconsistency	`step_adapter_restart`

5. Automated Test Verification Summary

Test Suite File	Scope & Assertions	Status
`Phase3KnowledgeBase.test.ts`	13 steps, 11 category roadmaps, 12 root causes, multi-OS commands	**7/7 PASSED**
`Phase3Interpreter.test.ts`	Multi-OS regex parsers (ipconfig, ping, nslookup, tracert, curl, wifi)	**21/21 PASSED**
`Phase2Classification.test.ts`	Problem classifier, typo tolerance, relevance, vague questions	**23/23 PASSED**
`Phase2UI.test.tsx`	Rejection UI, Follow-Up Card interaction	**3/3 PASSED**
`Phase1UI.test.tsx`	Clean Dashboard UI, Roadmap tracking, Step navigation, Modals	**6/6 PASSED**
Production Build (tsc -b)	TypeScript compilation & Vite bundle packaging	**0 ERRORS**